

LAL: Enhancing 3D Human Motion Prediction with Latency-aware Auxiliary Learning

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Abstract

Making accurate prediction of human motions based on the historical observation is a crucial technology for robots to collaborate with humans. Existing human motion prediction methods are all built under an ideal assumption that robots can instantaneously react, which ignores the time delay introduced during data processing & analysis and future reaction planning – jointly known as “response latency”. Consequently, the predictions made within this latency period become meaningless for practical use, as part of the time has passed and the corresponding real motions have already occurred before robot deliver its reaction. In this paper, we argue that the seemingly meaningless prediction period, however, can be leveraged to enhance prediction accuracy significantly. We propose LAL, a Latency-aware Auxiliary Learning framework, which shifts the existing “reaction instantaneous” convention into a new motion prediction paradigm with both latency compatibility and utility. The framework consists of two branches handling different tasks: the primary branch learns to directly predict the valid target (excluding the beginning latency period) based on observation; while the auxiliary branch learns the same target, but based on the reformed observation with additional latency data incorporated. A direct and effective way of auxiliary feature sharing is forced by our tailored consistency loss, to gradually integrate auxiliary latency insights into the primary prediction branch. Estimated feature statistics-based alignment method is presented as optional step for primary branch refinement. Experiments show that LAL achieves significant improvement on prediction accuracy, without additional time consumption during testing.

1. Introduction

3D human motion prediction is aimed at forecasting a future motion sequence based on historical observation, which has been widely used in many fields like autonomous driving [15, 36], motion tracking [8], and especially human-robot

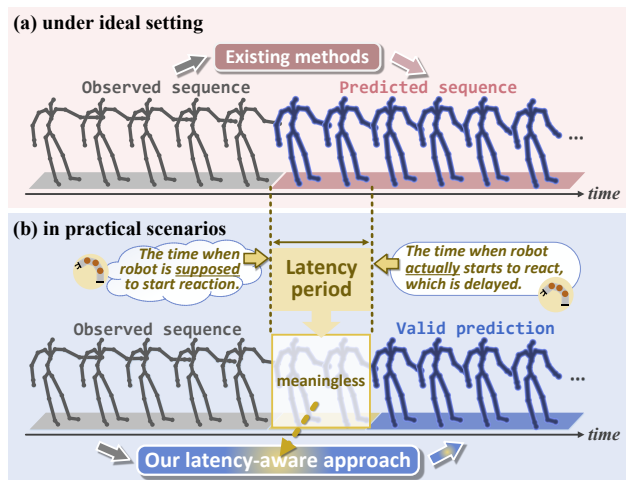


Figure 1. (a) Existing methods only focus on “what to predict”, while ideally overlook the latency issue lies in its applications – The reaction delivered by the robot is inevitably delayed, making it meaningless to predict motions within this period, because time is passing and real motions have occurred, while robot is still pending and have not decided its reaction yet. (b) We present a new latency-aware prediction paradigm, which turns the seemingly meaningless period into valuable auxiliary insights, to help improve prediction accuracy significantly.

collaboration [9, 19] where robots/machines are required to closely collaborate with humans.

Over the past years, research has focused on exploring spatio-temporal information of the observed motion sequence with deep learning-based methods, where typical architectures like RNNs [7, 11], GNNs [16] and GCNs [6, 24, 26] are employed, achieving great progress in the pursuit of prediction accuracy. Nevertheless, these methods forget the ultimate goal of human motion prediction, that is offering a foundation for robotic planning, to let robots generate proper reaction behaviors. Reaction often comes after several embedded steps that inevitably cost tens to hundreds of milliseconds [45]. Existing methods never consider such demand, and implicitly assume an ideal condition where

robot reaction can happen instantaneously without accounting for the actual time required.

Human motion prediction in practical applications is not as straightforward as in the above “reaction-instantaneous” methods. When a person moves, the intelligent system should (1) first detect and process the observed motion data, (2) then analyze them with the embedded algorithm to predict future motions; (3) make robotic planning and final execution. The entire process introduces time delay, which is known as response latency. As shown in Figure 1, when the system (e.g., robotic arm) enters into such period, the device is pending and does not engage in any physical behavior, while the human still continues to move. As a result, the delayed robot reaction cannot keep up with the ongoing human movements it is supposed to collaborate with, while the predictions made within this period is also outdated and become “meaningless”. Although, given this situation, an initial thought might be to remove latency period from the original prediction target, which enables supervision focus more on the latter valid region; our research shows that simply applying this operation to re-train state-of-the-art models does not yield satisfactory results.

We argue that the seemingly meaningless latency period, in contrast, can be transformed into a valuable element to enhance prediction accuracy. Instead of *ignoring* it (existing methods) or *discarding* it (the above naive solution), we integrate its details as part of observation to assist the predictive process. This is based on a simple and intuitive consensus that broader horizon of observation generally leads to better results. However, during testing, the system cannot obtain the human motion data of latency period in advance. Then we raise a question: How to utilize the latency insights but without directly accessing the latency data?

In this paper, we propose LAL, a novel framework that enables latency-aware prediction with auxiliary learning scheme. The framework consists of two branches dedicated to different tasks. The primary task skips the latency period and directly learn to predict the following valid target, based on the observed sequence. While for the auxiliary task, the motion data within latency period is also excluded from original target, but is integrated as supplement to construct the reformed observation, which allows the auxiliary branch to learn from a broader sequence and therefore achieve higher predictive accuracy. LAL differs from common practices of auxiliary learning that typically share encoding parameters across different tasks [18, 39]. Instead, to accommodate our distinct characteristics of inconsistent data dimensionality and high task similarity, we customize a feature consistency loss function that ensures effective sharing of auxiliary information through a gradual incorporation process. Additionally, we present an estimated feature statistics-based alignment method to further refine the primary branch, which performs auxiliary-feature summariza-

tion and primary-feature moment matching. This operation is optional and can be regarded as a post-regularization step, in order to strengthen the primary branch’s ability to better approximate the auxiliary latency feature distribution. During testing, only primary branch is executed, which predicts motions based on the original observation while benefiting from the latency insights, without any additional time cost.

In summary, our contributions are as follows:

- We are the first to introduce latency issue into human motion prediction, and argue to exploit this seemingly meaningless period to form a novel latency-aware prediction manner to suit practical use.
- We propose LAL, an auxiliary learning-based framework that enhances the primary prediction task with latency insights obtained from the auxiliary task, which is achieved by the customized consistency loss that ensures effective auxiliary information sharing. Estimated feature statistics alignment can be conducted to refine the primary branch.
- Experiments show that LAL enables existing response-instantaneous methods to support varying durations of latency, and significantly improve prediction performance.

2. Related Work

2.1. Human Motion Prediction

Based on a historical observed sequence, the goal of human motion prediction task is to accurately forecast the future motion sequence. As motion sequence is the time series of human poses, early research relies on sequential networks to predict motions frame by frame, such as RNNs [4, 11], LSTMs [7, 27] and GRUs [23, 29]. They suffer from error accumulation, a critical defect of recurrent network that may result in failure cases in convergence. Later, feed-forward networks have become the mainstream approach to avoid this defect, which predict all the poses (i.e., entire sequence) at once, such as CNNs and GANs. Recently, to fully explore the graph structure of human skeleton and extract spatial correlations among body joints, GNNs-based [16, 17] and GCNs-based [6, 22, 24–26, 30, 44] methods have emerged. Meanwhile, some research are proposed to handle spatio-temporal information, with architectures like transformer [1, 3] and MLP-mixer [2]. The new [38] reveals Euclidean geometric transformation principle in human motion, and present an efficient equivariant motion prediction model with invariant interaction reasoning. [34] considers the uncertainty problem and learns proper characteristics for future frames by a dynamic function. [35] first employs implicit neural representation into human motion prediction task. [42] addresses predictions under unexpected physical perturbation, or given unpremeditated external impact.

All of the above methods overlook the problem of response latency, that robots will inevitably delay and cannot keep up with the human motion it is supposed to inter-

act with, which makes the beginning part of the prediction meaningless. We, for the first time, reveal the importance of latency issue for human motion prediction in practical use, and present a novel latency-aware prediction manner.

2.2. Multi-Task Learning and Auxiliary Learning

Multi-task learning is aimed at improving the model performance and efficiency by simultaneously learning from multiple different tasks, with shared representation and information across all tasks at early layers of the network [18, 43]. By selecting one primary task and others as auxiliary task(s), auxiliary learning approach is formed, where the primary task can be assisted by auxiliary task information, to enable regularized learning process, richer feature learning, knowledge transfer, or increased robustness. So far, this idea has been applied in many fields like object detection [14, 21], semantic segmentation [39], speech recognition [32], human mobility prediction [40], future depth prediction for videos [20], and continual learning [12], etc.

Auxiliary Learning in Human Motion Prediction. Particularly, up to now, only two works have considered auxiliary learning in human motion prediction [5, 37]. Both of them build the network by employing two motion-related tasks (such as motion denoising and motion repairing) as auxiliary branches to assist the primary prediction task, where encoding parameters are shared among all the tasks to enrich the primary information. However, our latency-aware auxiliary learning scheme is completely different from the above methods, owing to its high task similarity and inconsistent data dimensionality between the primary and auxiliary task (may refer to Section 3.2 *Training*), and we design a special loss function tailored to this situation to ensure effective feature sharing.

3. Proposed Method

In this section, we formally describe human motion prediction task with the consideration of latency issue. Then, we present our latency-aware framework with auxiliary learning scheme, and provide detailed discussion.

3.1. Formulation

Preliminary. Existing works for human motion prediction follow the same task definition and training procedure. Suppose the observed motion sequence $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]$, this task aims to accurately predict the subsequent T poses as $\hat{\mathbf{Y}} = [\hat{\mathbf{y}}_1, \hat{\mathbf{y}}_2, \dots, \hat{\mathbf{y}}_T]$, with the corresponding ground truth denoted as $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_T]$. Each human pose is represented by 3D skeletal joint coordinate, i.e., $\mathbf{x}_i, \hat{\mathbf{y}}_i, \mathbf{y}_i \in \mathbb{R}^{J \times 3}$, where J is the number of joints. N and T are the pose number of the observed sequence and predictive sequence target. These methods ignore the latency problem in practical applications, and only focus on pursuing high prediction accuracy theoretically.

Setup for Latency-aware Prediction. To enable latency compatibility and utility, we reform motion prediction task as below. We define the motion sequence that falls into latency period as $\mathbf{Y}^{lat} = [\mathbf{y}_1, \dots, \mathbf{y}_{T_L}]$, with T_L denoted latency length, and we cut it from the original prediction target $\mathbf{Y}$ to form the new, valid target $\mathbf{Y}^{val} = [\mathbf{y}_{T_L+1}, \dots, \mathbf{y}_T]$. Our goal is to skip $\mathbf{Y}^{lat}$ and directly predict the valid $\mathbf{Y}^{val}$ based on $\mathbf{X}$, but with the assistance of auxiliary information learned from $\mathbf{Y}^{lat}$. Usually T_L is set small, just a few poses, whose duration matches the common latency period (human motion framerate is around 25 fps, i.e., each pose interval equals to 40 milliseconds).

3.2. Latency-aware Auxiliary Learning

Motivation. While existing response-instantaneous methods predict the original $\hat{\mathbf{Y}} = [\hat{\mathbf{Y}}^{lat}, \hat{\mathbf{Y}}^{val}]$ that contains the meaningless $\hat{\mathbf{Y}}^{lat}$ period, the initial thought is to avoid wasting training resources over such invalid target, that is discarding $\mathbf{Y}^{lat}$ and solely learning to predict $\mathbf{Y}^{val}$. However, in Figure 2 (left), the naive operation shows no improvement. This is natural, as the extrapolation ability of the same model on the same input at fixed positions are relatively stable or “limited”. In contrast, we notice that the prediction performance can be significantly improved if we merge latency data as part of the observation (right). This motivates us to further exploit the latency information from $\mathbf{Y}^{lat}$ as auxiliary supplement for original input to assist the prediction process, which makes the seemingly meaningless motion data useful.

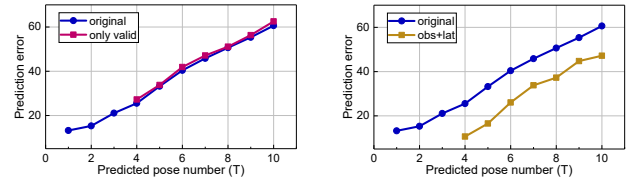


Figure 2. Suppose a motion sample with $N = 10$ poses as observation and $T = 10$ as target, wherein the first 3 poses as latency region ($T_L = 3$). *Original*: standard setting that predicts 10 poses based on 10-pose observation; *Only valid*: remove the supervision over latency and directly learn to predict the valid target (no improvement); *Obs+lat*: treat latency data as additional observation, i.e., 13 poses of observation (significant progress).

Overview. Our Latency-aware Auxiliary Learning (LAL) framework consists of two branches dedicated to primary and auxiliary tasks. The primary branch is to directly predict the valid target based on the observation, i.e., $\hat{\mathbf{Y}}_{Pri}^{val} = f_{\mathcal{P}}(\mathbf{X})$. The auxiliary branch is also aimed at predicting the valid target, but based on the reformed observation that integrates latency motion data, denoted as $\hat{\mathbf{Y}}_{Aux}^{val} = f_{\mathcal{A}}(\mathbf{X}, \mathbf{Y}^{lat})$. To let the auxiliary latency information shared by the primary prediction, we tailor a loss function that forces feature consistency between the encoded latent

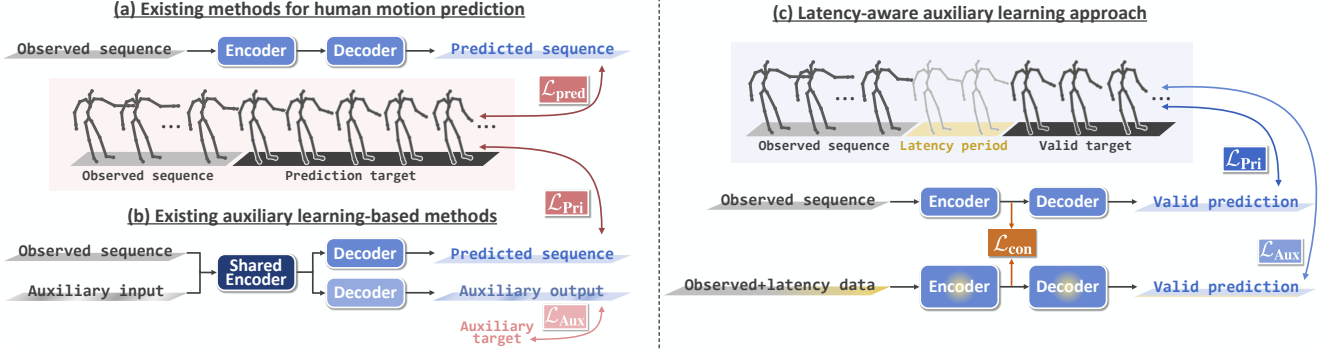


Figure 3. (a) Standard human motion prediction methods; (b) existing auxiliary learning-based methods; (c) our latency-aware auxiliary learning. Both (a) and (b) are under the reaction-instantaneous assumption. Particularly, the existing two works of auxiliary learning for motion prediction [5, 37] follow the same pattern, that is sharing encoding parameters across the primary motion prediction task and the auxiliary tasks (e.g., motion repairing/denoising). Differently, ♦ we claim the latency issue, and propose to transform the useless latency data into valuable additional input for the auxiliary prediction task, which benefits the primary prediction process with latency insights; ♦ to accommodate high task similarity and inconsistent input dimensionality, we customize a new pattern of auxiliary learning that employs separate pipelines for each task, with $\mathcal{L}_{\text{con}}$ aligning their respective encoded latent representations to realize effective feature sharing.

representations of each branch, so that the latency information can gradually and implicitly flow into the primary branch during training. In testing phase, where only the primary branch is executed without pre-access to latency motion data, the primary prediction can still be enhanced significantly owing to the acquired auxiliary latency insights.

♦ **Architecture and Training.** We present the backbone of our LAL framework in Figure 3, which is constructed with a very simple yet effective way. As LAL is used to enable many existing methods to realize latency-aware prediction, each branch is structurally consistent with the specific model it applies to. Concretely, both the encoding and decoding parts in each branch mirror the corresponding modules of the specific model, except the input and output layers of the auxiliary branch that handle different data dimensions. Note that this only suits to those motion prediction models with mainstream encoder-decoder architecture.

To facilitate effective sharing of auxiliary latency information, we specially design a feature consistency loss function that forces the encoded latent representations of the two branches to converge as closely as possible. This feature alignment strategy enables the gradual integration of auxiliary insights into the primary prediction process. We formulate this loss as:

$$\mathcal{L}_{\text{con}} = \|\text{Enc}_{\mathcal{P}}(\mathbf{X}) - \text{Enc}_{\mathcal{A}}(\mathbf{X}, \mathbf{Y}^{\text{lat}})\|_2, \quad (1)$$

where $\text{Enc}_{\mathcal{P}}(\cdot)$ and $\text{Enc}_{\mathcal{A}}(\cdot)$ denote the encoded representation in the primary and auxiliary branch, respectively.

Notably, we are different from the common practice in auxiliary learning that directly share the encoding parameters for all tasks. Instead, we use $\mathcal{L}_{\text{con}}$ loss to enrich feature representation for primary prediction branch. The reasons are twofolds: First, the two tasks in LAL have different in-

puts $\mathbf{X}$ and $(\mathbf{X}, \mathbf{Y}^{\text{lat}})$ with inconsistent data size. Simply inserting a linear layer, or temporal convolutional layer [41] to align data dimensions would risk losing its valuable information, as human motion data is delicate and highly complex, which is hard to retain with naive one-layer operation.

Second, existing auxiliary learning approaches typically share encoding network across different *types* of task to enhance the model representation capacity and robustness, like the primary motion prediction task and auxiliary motion denoising task in [37]. However, the two tasks in our LAL essentially are of the same nature (both are motion prediction tasks), and with similar inputs (only a few more frames in auxiliary branch). If we employ the common strategy like [37], it would be challenging to distinguish their subtle yet valuable differences of motion information, and cannot fully exploit the advantages of each task. Therefore, we propose to enable feature consistency with $\mathcal{L}_{\text{con}}$ loss at the end of encoding as a more direct and strong force. In Section 4.4.1, we also conduct experiments to validate the above discussion.

We follow current works [17, 25, 38] to employ Mean Per Joint Position Error (MPJPE) to measure the prediction accuracy in each task, but differently, we calculate this loss over our valid $(T - T_L)$ -frame horizon, rather than original T -frame one. The prediction loss on primary branch can be expressed as:

$$\mathcal{L}_{\text{Pri}} = \frac{1}{T - T_L} \sum_{t=T_L+1}^T \|\hat{\mathbf{y}}_{\text{Pri},t} - \mathbf{y}_t\|_2^2, \quad (2)$$

and the prediction on auxiliary branch as:

$$\mathcal{L}_{\text{Aux}} = \frac{1}{T - T_L} \sum_{t=T_L+1}^T \|\hat{\mathbf{y}}_{\text{Aux},t} - \mathbf{y}_t\|_2^2, \quad (3)$$

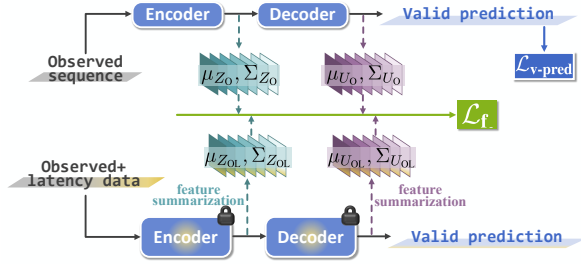


Figure 4. Refinement stage for LAL. With the parameters of auxiliary branch fixed, we summarize the observation-latency feature distributions after the encoder and decoder with first and second-order moments. The primary branch can be further refined through feature alignment of each iteration, to let the primary branch learn to produce the auxiliary distribution with only observation data.

where $\hat{\mathbf{y}}_{\text{Pri},t}$ and $\hat{\mathbf{y}}_{\text{Aux},t}$ denote the primary and auxiliary predicted pose at frame t , while $\mathbf{y}_t$ denotes the corresponding ground truth.

Finally, based on the above, the total training loss can be expressed as:

$$\mathcal{L}_{\text{LAL}} = \lambda_{\mathcal{P}} \mathcal{L}_{\text{Pri}} + \lambda_{\mathcal{A}} \mathcal{L}_{\text{Aux}} + \mathcal{L}_{\text{con}}, \quad (4)$$

where $\lambda_{\mathcal{P}}$ and $\mu_{\mathcal{A}}$ are the relative weights of each loss component. In testing phase, we only need to perform the primary prediction branch to achieve prediction with latency compatibility and utility.

3.3. Refinement for LAL

The above discussed LAL provides an end-to-end training strategy and improve the prediction performance with only one stage of training. In this section, we present an optional refinement step for further feature alignment. Recall $\mathcal{L}_{\text{con}}$ that enables balanced parallel training to realize auxiliary feature shared by the primary branch. Inspired by [31], we specify that using the square distance of the first and second moments between two feature distributions, could express the discrepancy of features with and without latency data, which allows the primary branch to further approximate the auxiliary latency branch. To be specific, when the training of LAL finishes, we fix the parameters of auxiliary branch, and summarize the encoded feature of motion set $Z = \{z_1^T, z_2^T, \dots, z_M^T\}$ by calculating the mean $\mu_Z = \frac{1}{M} \sum_m^M z_m$, and covariance matrix $\Sigma_Z = \frac{1}{M-1} (Z^T Z - \frac{1}{M} (I^T Z)^T (I^T Z))$, where M denotes sample number. We can regard the former operation as channel-wise batch normalization statistics while the latter is more efficient and informative. For clearer representation, we denote the encoded latent statistics of observation-latency data (i.e., OL) at the auxiliary branch as $(\mu_{Z_{\text{OL}}}, \Sigma_{Z_{\text{OL}}})$. Similarly, the encoded latent statistics of observation-only data (i.e., O) at the primary branch is denoted as $(\mu_{Z_{\text{O}}}, \Sigma_{Z_{\text{O}}})$,

and we minimize the distance between them and the pre-summarized auxiliary statistics (i.e., $(\mu_{Z_{\text{OL}}}, \Sigma_{Z_{\text{OL}}})$) to further refine the parameters of the primary branch, with the loss expressed as:

$$\mathcal{L}_{\text{fea},Z} = \|\mu_{Z_{\text{OL}}} - \mu_{Z_{\text{O}}}\|_2^2 + \|\Sigma_{Z_{\text{OL}}} - \Sigma_{Z_{\text{O}}}\|_F^2. \quad (5)$$

Same for the decoded statistics $U = \{u_1^T, u_2^T, \dots, u_M^T\}$, we represent the estimated statistics at the auxiliary branch and the primary branch as $(\mu_{U_{\text{OL}}}, \Sigma_{U_{\text{OL}}})$ and $(\mu_{U_{\text{O}}}, \Sigma_{U_{\text{O}}})$, respectively. The loss can therefore be denoted as $\mathcal{L}_{\text{fea},U}$ similar to Eq. (5). The final alignment loss for refinement stage can be formulated as:

$$\mathcal{L}_{\text{refine}} = \mathcal{L}_{\text{v-pred}} + \lambda_Z \mathcal{L}_{\text{fea},Z} + \lambda_U \mathcal{L}_{\text{fea},U}, \quad (6)$$

where $\mathcal{L}_{\text{v-pred}}$ is the prediction loss of the valid region on the primary branch, λ_Z and λ_U are hyper-parameters that control the importance of each component. Note that this step is not mandatory, as the two-stage structure may sacrifice part of the training simplicity of our LAL, but obtain better prediction performance.

4. Experiments

In this section, we evaluate our LAL approach for human motion prediction with latency-aware prediction manner. Three representative baselines are selected to be re-built with our LAL framework, and we provide extensive numerical results with visualizations, followed by ablation study.

4.1. Datasets & Evaluation Metric

Human3.6M [10] is one of the most widely used datasets in human motion-related tasks, which involves seven actors performing 15 categories of actions, such as *walking*, *eating* and *smoking*, etc. Motions are captured and down-sampled into sequence data of 25 *fps*. Each pose is depicted by 3D coordinates of 32 body joints. Following existing [17, 25, 30], we consider 22 of them; and use Subjects 1, 6, 7, 8, 9 for training, S5 for testing and S11 for validation.

CMU-Mocap contains eight actions (such as *basketball*, *directing traffic*, *jumping*, etc.) denoted by 28 joints of 3D coordinates. We follow [25] to leave 25 joints for experiments. The frame rate is also 25 *fps*. The training and testing splits are also the same as [25].

3DPW (The 3D Pose in the Wild) [33] dataset is more challenging that contains more than 51k frames of indoor or outdoor human motion activities, captured by a moving camera with 30 *fps*. Each body is depicted by 3D coordinates of 23 joints. We also use the same training, testing and validation sets as existing [17, 25] to ensure fair comparison.

Evaluate Metric. Following current human motion prediction works [17, 25, 30, 38], we evaluate our approach by Mean Per Joint Position Error (MPJPE), which calculates the average L_2 distance between the predicted joints and target ones at all timestamps.

millisecond	smoking				directions				sitting				waiting			
	80	160	320	400	80	160	320	400	80	160	320	400	80	160	320	400
Res. sup [27]	23.0	42.6	70.1	82.7	35.4	57.3	76.3	87.7	42.6	81.4	134.7	151.8	30.6	57.8	106.2	121.5
DMGNN [16]	9.0	17.6	32.1	40.3	13.1	24.6	64.7	81.9	11.9	25.1	44.6	50.2	12.2	24.2	59.6	77.5
PGBIG [24]	6.6	14.1	28.2	34.7	7.2	17.6	40.9	51.5	8.8	19.2	42.4	53.8	8.9	20.1	43.6	54.3
LTD [25]	8.0	16.2	31.9	38.9	9.0	19.9	43.4	53.7	10.6	21.9	46.3	57.9	11.4	24.0	50.1	61.5
LTD-LAL-3	-	14.5	29.2	36.0	-	17.5	40.1	50.4	-	20.0	43.7	55.1	-	20.3	44.5	55.7
STS [30]	13.1	20.2	37.7	44.7	14.2	24.3	44.2	53.2	15.2	23.0	46.8	58.3	16.3	27.3	48.1	59.8
STS-LAL-3	-	22.6	31.0	35.3	-	20.5	32.2	44.8	-	20.4	32.8	44.6	-	26.5	36.9	47.0
SPGSN [17]	6.7	13.8	28.0	34.6	7.4	17.2	39.8	50.3	9.3	19.4	42.3	53.6	9.2	19.8	43.1	54.1
SPGSN-LAL-3	-	12.4	26.6	31.9	-	16.1	36.6	46.4	-	19.2	40.4	51.2	-	18.0	39.8	49.9

Table 1. Comparisons of MPJPE errors on 4 typical activities in Human3.6M. We set $T_L = 3$ (i.e., 120 ms), indicating the number of poses that fall into latency period. Lower errors are in **bold**, which show significant improvement with LAL.

4.2. Baselines

We provide extensive comparisons with existing methods Res. sup [27], DMGNN [16], LTD [25], STS [30], PGBIG [24] and SPGSN [17]. We select three representative baselines as below, and transform them into latency-aware prediction manner to validate the effectiveness of LAL.

LTD is the first to introduce GCN into motion prediction, which employs 12 GCN blocks with skip connections to depict spatial correlations of body joints. It also uses Discrete Cosine Transformation to extract temporal information.

STS proposes a single-graph structure to enable cross-talk between spatial and temporal domains, which separates the graph into space and time matrices to obtain joint-joint and time-time interactions.

SPGSN develops adaptive graph scattering to capture detailed spatial and spectral features, with trainable band-pass filters to decompose features into broader spectrum bands. It divides human skeleton into body-parts to extract diverse motion dynamics, and fuse them for full-body encoding.

Implementation Details. We conduct our experiments under Pytorch [28] with Adam optimizer [13] on a single NVIDIA RTX 3080. We train the above three baselines with LAL for 50 epochs. The learning rate for LTD-LAL, STS-LAL and SPGSN-LAL is 0.001, 0.01 and 0.001, respectively. The settings of learning rate decay and batch size are consistent with the original baselines. For main-stream reaction-instantaneous methods, short-term prediction indicates observation of $N = 10$ and prediction target $T = 10$ (400 ms), while long-term prediction indicates $N = 10$ and $T = 25$ (1000 ms). Differently, as our LAL address the latency issue, the beginning few timestamps of baseline-LAL are skipped as invalid, and therefore we do not report their prediction error.

4.3. Results

Human3.6M. First, we use four activities as examples to explain our presentation of short-term prediction errors in

millisecond	average						running time (ms)
	80	160	320	400	560	1000	
LTD [25]	12.7	26.1	52.3	63.5	81.6	114.3	36.87
LTD-LAL-1	12.1	25.4	51.9	63.0	81.8	114.5	36.84
LTD-LAL-2	-	24.6	50.4	62.4	81.0	113.5	36.77
LTD-LAL-3	-	23.1	48.3	59.6	80.4	111.7	36.76
LTD-LAL-4	-	-	47.9	59.2	79.6	110.1	36.71
LTD-LAL-5	-	-	47.0	58.7	77.1	108.5	36.58
STS [30]	15.3	25.5	50.6	60.6	80.7	113.3	28.20
STS-LAL-1	15.8	25.8	50.4	60.1	79.8	111.9	28.18
STS-LAL-2	-	25.7	45.1	58.3	78.1	111.7	28.11
STS-LAL-3	-	25.2	41.8	51.4	79.6	110.5	28.09
STS-LAL-4	-	-	39.5	51.8	75.3	109.6	28.03
STS-LAL-5	-	-	37.9	49.2	71.3	106.6	27.98
SPGSN [17]	10.5	22.3	47.0	58.2	77.4	109.6	45.12
SPGSN-LAL-2	-	21.7	45.7	56.0	75.3	109.1	45.08
SPGSN-LAL-3	-	20.8	44.1	54.6	74.8	108.6	45.10
SPGSN-LAL-4	-	-	43.6	54.1	73.9	107.3	45.04
SPGSN-LAL-5	-	-	42.8	53.0	72.5	104.1	45.03

Table 2. Comparisons of MPJPE average errors in Human3.6M, where $T_L = 1, 2, 3, 4, 5$, supporting latency duration from ≤ 40 ms to ≤ 200 ms. LAL helps achieve improved performance without additional time cost during testing, shown in the right column.

Table 1. Baseline-LAL- T_L indicate the results of each baseline under LAL framework, highlighted by background color. In this table we set $T_L = 3$, which means the latency occupies the duration of 3 poses (equivalent to 120 ms, as the motion sampling rate is 25 *fps*). Therefore, the original 80 ms timestamps become invalid, denoted as “-”. Our LAL approach can benefit existing methods in most cases.

In Table 2, we further demonstrate the average performance of short-term and long-term prediction errors when $T_L = 1, 2, 3, 4, 5$, with the running time comparison. From the table, our effective utility of latency insights indeed

millisecond	CMU-Mocap			3DPW		
	160	400	1000	200	500	1000
LTD [25]	18.0	41.0	81.9	35.6	80.2	117.8
LTD-LAL-1	17.4	39.8	81.8	N/A	N/A	N/A
LTD-LAL-2	16.7	39.2	80.7	34.3	78.8	116.4
LTD-LAL-3	15.6	38.5	79.7	33.1	77.0	115.3
LTD-LAL-4	-	38.3	78.4	31.7	76.8	113.7
LTD-LAL-5	-	37.8	76.5	31.2	75.9	110.8
STS [30]	18.2	41.1	81.8	37.8	77.3	112.2
STS-LAL-1	17.9	41.4	81.3	37.2	76.5	111.4
STS-LAL-2	16.8	39.5	80.7	36.0	76.4	110.8
STS-LAL-3	15.3	37.2	79.1	33.8	75.3	109.0
STS-LAL-4	-	37.6	76.2	33.4	74.7	107.6
STS-LAL-5	-	36.9	75.8	34.6	73.1	106.6
SPGSN [17]	14.8	37.0	77.8	32.9	76.2	111.1
SPGSN-LAL-2	13.4	35.7	76.8	30.9	75.6	109.3
SPGSN-LAL-3	12.5	34.9	75.1	30.6	74.3	107.8
SPGSN-LAL-4	-	34.8	74.9	28.4	72.8	106.9
SPGSN-LAL-5	-	33.9	74.2	27.1	71.3	106.3

Table 3. Comparisons of average MPJPE errors in CMU-Mocap and 3DPW, with $T_L = 1, 2, 3, 4, 5$, supporting latency duration from ≤ 40 ms to ≤ 200 ms for CMU-Mocap, from ≤ 33.4 ms to ≤ 166.5 ms for 3DPW (due to different frame rates). Our LAL constantly yield improved prediction performance.

helps improve the prediction over the valid target region, without introducing additional time cost during testing.

Notably, our methods show constantly promising results regardless of the specific latency duration. This capability is particularly beneficial for real-world deployments, where devices of different performance levels may be used, ranging from lightweight or low-capacity devices to more powerful ones. By pre-determining the latency-related indicators of the given device, we could select the corresponding latency configuration of our LAL accordingly, and therefore employ the best-performing model for specific scenarios.

CMU and 3DPW. We provide the average results on this two datasets in Table 3, which also demonstrate significant improvement achieved by LAL-assisted models. Due to space limit, we omit the comparison for running time, but the trend appears consistent with the results in Table 2. The entire table may refer to our supplementary material.

♦ **Sanity Check.** As we have mentioned, T_L after each “baseline-LAL” means the number of poses that the latency period occupies, which indicates the *maximum* latency duration it can tolerate, and we should ensure *practical latency duration* does not exceed *this maximum*. This is why there is no SPGSN-LAL-1 in Table 2, 3, and no LTD-LAL-1 for 3DPW in Table 3. The time cost for them to calculate forecast content is already larger than $T_L = 1$ duration (40 ms), let alone that we have not yet included other time cost required for data processing and future reaction planning,

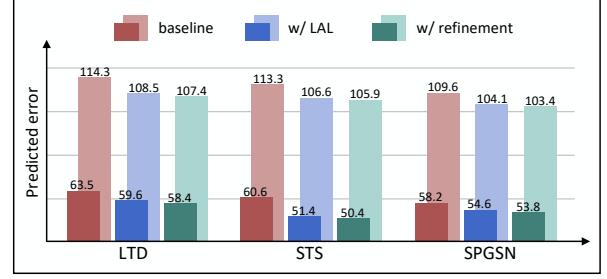


Figure 5. Comparisons of prediction errors among baselines, with LAL, and with optional refinement step on Human3.6M. Dark colors indicate errors at 400 ms, while faded colors 1000 ms.

which also account for total latency in practice.

On the other hand, we do not further increase T_L , as it may be too low-efficient for practical use if the latency period lasts for more than 25% of the valid prediction target.

♦ **Fairness discussion.** Notably, although we integrate additional motion data during training, we still make predictions *only* based on original observation in testing phase, so the comparison between baselines without/with LAL is fair.

Refinement for LAL. In Figure 5, we provide the results of baselines built with refined LAL on Human3.6M. For each dark-faded bar pair, the dark indicates errors at 400 ms while faded indicates 1000 ms. We employ refined/non-refined LAL with $T_L = 3$ to produce the results at 400 ms, and $T_L = 5$ at 1000 ms. Our feature statistics-based refinement strategy for the primary prediction branch can further bring benefits to prediction accuracy from the figure.

Visualization is provided in Figure 6 to present samples of *walking* and *wash window*. The predictive discrepancies are marked in red boxes, which are alleviated by LAL design.

4.4. Ablation Study

4.4.1. Validating the effectiveness of LAL

Why not discard/reduce the supervision over latency period. As we mentioned in Section 3.2 *Motivation*, the initial thought to treat invalid target is directly discarding it, or reducing the supervision over it, to let the model focus more on the valid target. Taking $T_L = 3$ and STS baseline as example, the original target length is $T = 10$, but the model is trained with the supervision of $\mathcal{L} = \theta \mathcal{L}_{\text{lat}} + \mathcal{L}_{\text{val}}, \theta \in (0, 1)$ to control the reduction of the prediction loss within latency period. Figure 7 (left) shows results when $\theta = 0.2, 0.5, 0.8$ respectively. We also include *only valid* (i.e., no $\mathcal{L}_{\text{lat}}$). From the figure, the naive strategy does not bring much benefits. However, our LAL exhibits significant improvement, which **validates** its effective utility of latency information, and proves the soundness of our auxiliary learning scheme.

Why not use shared encoder. We follow the common practice in auxiliary learning to use shared encoder across different tasks. As our inputs $\mathbf{X}$ and $(\mathbf{X}, \mathbf{Y}^{\text{lat}})$ have inconsistent

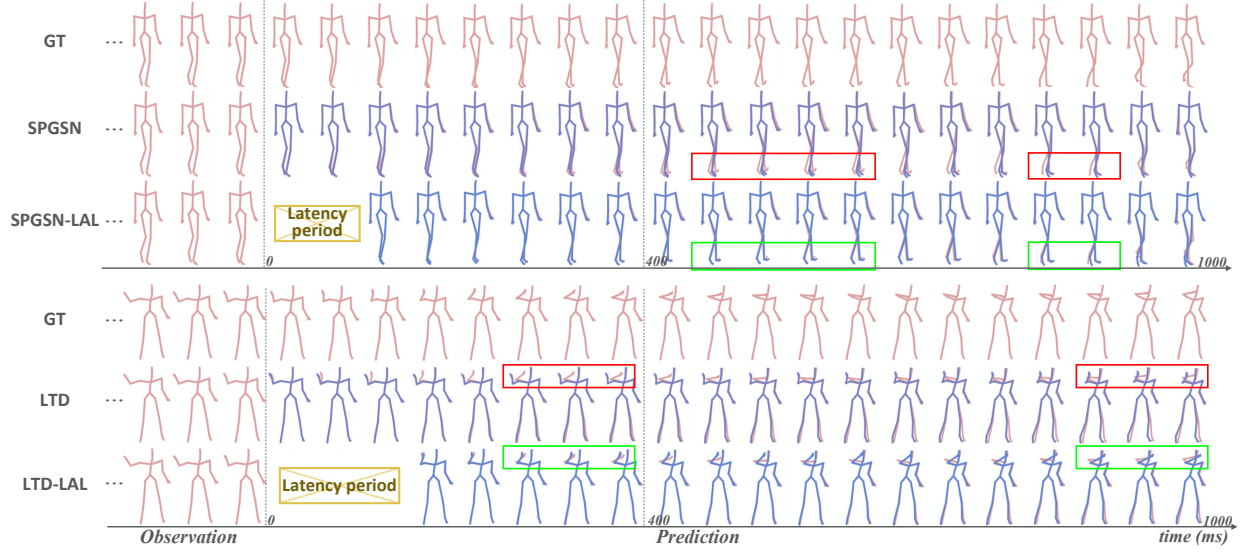


Figure 6. Visualized comparison between baselines with and without LAL assisted. Pink skeletons indicate observation or ground truth, and other colors indicate predictions. We set $T_L = 3$ in the upper case and $T_L = 4$ in the lower case as latency period (Note that we only cover 2 and 3 poses in the figure, as we proportionally reduce the number of displayed poses due to space limit). Highlighted in green boxes, our predictions are closer to GT than existing methods, owing to the effective utility of auxiliary latency insights.

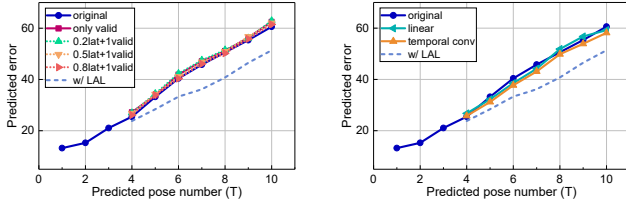


Figure 7. Blue solid line as baseline, and blue dashed line as with LAL. Left: “ $\theta_{lat}+1valid$ ” indicate the reduced intensity of supervision over the prediction within latency period (i.e., invalid part). Right: share encoding between the two tasks, with linear or temporal convolutional operation for dimension alignment.

data size, we add another layer to align the dimension. We employ linear operation or temporal convolutional [41] operation before the shared encoder, and results of each design is shown in Figure 7 (right), where only limited improvement is achieved. In Section 3.2 Training, we have fully discussed its reasons, that shared encoding cannot handle our distinct characteristics (i.e., inconsistent data dimensionality and high task similarity). Our LAL, in contrast, achieves great progress owing to the specially designed loss $\mathcal{L}_{con}$, enabling direct yet strong auxiliary information sharing. The discrepancy in the figure exactly **validates** that our separate encoding with feature consistency loss are effective.

4.4.2. Ablation on training loss hyper-parameters

In Figure 8, we show the results of different (λ_P, λ_A) weights in our total loss function Eq. (4). We use STS-LAL with $T_L = 3$ as example, where the overall best per-

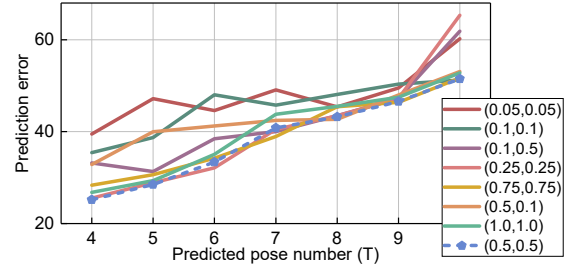


Figure 8. Results of different (λ_P, λ_A) settings on Eq (4). The best is when $\lambda_P = 0.5, \lambda_A = 0.5$, drawn in dashed line.

formance is achieved when $(\lambda_P = 0.5, \lambda_A = 0.5)$. This setting may vary for different baselines, and we will provide detailed ablations in supply. Ablations for each term in refinement loss will also be included.

5. Conclusion

We are the first to introduce the latency issue into human motion prediction, and argue to turn the latency data into valuable supplementary latency insights. We propose LAL, a latency-aware auxiliary learning framework, which enhances the primary motion prediction task by integrating latency information from an auxiliary prediction branch, and leverages a tailored consistency loss to ensure effective feature sharing, which, for the first time, enables a new prediction manner with latency compatibility and utility. Experiments show that LAL can be simply applied to many existing models, to improve prediction accuracy significantly.

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